

Plagiarism and Proper Reference

...

Lecture 2

Steps for Generating References Section

- ❑ Create a .bib file in the same directory as your .tex file. (eg., *ref.bib*)
- ❑ Copy and paste the BibTeX entry (found under “Cite”) from the literature you want to reference—[Google Scholar](#) is a good source. (see [slide 5](#))
- ❑ Choose correct citation among MLA, [APA](#), IEEE... (see [slide 7](#)).
- ❑ Master these four basic commands from page (see [slide 6](#))



 main.tex

 ref.bib

This is worth reviewing → [Bibtex.eu](#)

Building Your Bibliography File (.bib) the Easy Way

Google Scholar

Linear search time complexity

Articles

About 6,970,000 results (0.11 sec)

Any time

Since 2025

Since 2024

Since 2021

Custom range...

Sort by relevance

Sort by date

Any type

Review articles

☐ include patents

☒ include citations

☒ Create alert

Complexity analysis of second-order line-search algorithms for smooth nonconvex optimization [PDF] siam.org

CW Royer, SJ Wright - SIAM Journal on Optimization, 2018 - SIAM

... of a **search** direction, followed by a backtracking **line search** ... the **search** directions, using iterative methods in **linear** algebra... We derive modified convergence and **complexity** results for ...

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Linear bandit algorithms with sublinear time complexity [PDF] mlr.press

S Yang, T Ren, S Shakkottai, E Price... - International ..., 2022 - proceedings.mlr.press

... We propose two **linear** bandits algorithms with per-step **complexity** sublinear in the number of ... Compared with the **linear time complexity** baselines, our algorithms can offer a 72 times ...

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L* algorithm—A linear computational complexity graph searching algorithm for path planning [PDF] springer.com

A Niewola, L Podsedkowski - Journal of Intelligent & Robotic Systems, 2018 - Springer

... (buckets) ensures the **linear computational complexity** of the L* ... the optimality and **linear computational complexity** with the ... a method of estimating the execution **time** of the A* and the L...

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Complexity analysis of second-order line-search algorithms for smooth nonconvex optimization

CW Royer, SJ Wright - SIAM Journal on Optimization, 2018 - SIAM

... of a **search** direction, followed by a backtracking **line search** ... the **search** directions, using iterative methods in **linear** algebra... We derive modified convergence and **complexity** results for ...

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Cite

MLA Royer, Clément W., and Stephen J. Wright. "Complexity analysis of second-order line-search algorithms for smooth nonconvex optimization." *SIAM Journal on Optimization* 28.2 (2018): 1448-1477.

APA Royer, C. W., & Wright, S. J. (2018). Complexity analysis of second-order line-search algorithms for smooth nonconvex optimization. *SIAM Journal on Optimization*, 28(2), 1448-1477.

Chicago Royer, Clément W., and Stephen J. Wright. "Complexity analysis of second-order line-search algorithms for smooth nonconvex optimization." *SIAM Journal on Optimization* 28, no. 2 (2018): 1448-1477.

Harvard Royer, C.W. and Wright, S.J., 2018. Complexity analysis of second-order line-search algorithms for smooth nonconvex optimization. *SIAM Journal on Optimization*, 28(2), pp.1448-1477.

Vancouver Royer CW, Wright SJ. Complexity analysis of second-order line-search algorithms for smooth nonconvex optimization. *SIAM Journal on Optimization*. 2018;28(2):1448-77.

BibTeX EndNote RefMan RefWorks

```
@article{royer2018complexity,
  title={Complexity analysis of second-order line-search
  author={Royer, Cl(\e)ment W and Wright, Stephen J},
  journal={SIAM Journal on Optimization},
  volume={28},
  number={2},
  pages={1448--1477},
  year={2018},
  publisher={SIAM}
}
```

Fundamental Commands for Bibliography and Citation

 `\usepackage{biblatex}`

Loads the citation package and allows advanced referencing (APA, MLA, etc.).

 `\addbibresource{filename.bib}`

Links your .bib file to the LaTeX document (e.g., ref.bib).

 `\cite{key}`

Inserts an in-text citation for the source identified by its BibTeX key (e.g. `\cite{author}`).

 `\printbibliography`

Generates the full bibliography/reference list at the end of your document.

APA, MLA, IEEE citation

- To change your citation format to **APA**, add `[style = apa]` to your biblatex preamble.
 - Example code: `\usepackage[style=apa]{biblatex}`
- To use **MLA** format, add `[style = mla]` instead..
 - Example code: `\usepackage[style=mla]{biblatex}`
- To use **IEEE**, add `[style = ieee]` instead.
 - Example code: `\usepackage[style=ieee]{biblatex}`



Key point:

The `style=apa`, `mla`, `ieee`, etc. options vary by citation style, but the rest of your LaTeX code remains the same.

Remember:

LaTeX can change the entire pdf with one word.

Working Example: $\LaTeX$ + Bib $\LaTeX$ Citation

```
\documentclass[12pt]{report}
\usepackage[style=apa]{biblatex} % Initializes biblatex with APA citation style
\addbibresource{ref.bib} %Imports the bibliography file

\begin{document}

Digital literacy has become essential in today's world, and web design plays a significant
role in that landscape. As noted by Castro, understanding HTML is foundational to navigating
and shaping the internet (\cite{castro2003html}). % Best practice: paraphrase and cite using
(\cite{author})—avoid direct copy-paste.

Cultural practices are equally important in understanding societies. Traditional fishing
methods, for example, offer insight into human-environment relationships, especially in
places like Japan (\cite{japan}).

Furthermore, visual media such as manga contributes to the cultural narrative of Japan. Ito
explores how manga reflects and shapes historical consciousness (\cite{ito2014manga}).

\printbibliography % Renders bibliography in APA style using entries from the .bib file
\end{document}
```

Fig 1: main.tex file

```
@book{castro2003html,
  title={HTML for the world wide web},
  author={Castro, Elizabeth},
  year={2003},
  publisher={Peachpit Press}
}

@article{japan,
  title={Anthropology of fishing},
  author={Acheson, James M},
  journal={Annual review of anthropology},
  volume={10},
  pages={275--316},
  year={1981},
  publisher={JSTOR}
}

@incollection{ito2014manga,
  title={Manga in Japanese history},
  author={Ito, Kinko},
  booktitle={Japanese visual culture},
  pages={26--47},
  year={2014},
  publisher={Routledge}
}
```

Fig 2: ref.bib file

Document sectioning

-1	<code>\part{part}</code>
0	<code>\chapter{chapter}</code>
1	<code>\section{section}</code>
2	<code>\subsection{subsection}</code>
3	<code>\subsubsection{subsubsection}</code>
4	<code>\paragraph{paragraph}</code>
5	<code>\subparagraph{subparagraph}</code>

- Usually, `\section` is the top-level document command in most documents. eg: article.
- However, in **reports** or books, and similar long documents, this would be `\chapter` or `\part`.

Document section: `\chapter{}`

1. Use this to create project reports!
2. Can only be used in report and book class.
3. <https://www.overleaf.com/4249489998dgznvxfxmgpr#4e838e>

```
\documentclass{report}
\title{Sections and Chapters}
\author{Danish Puri}
\date{\today}
\begin{document}
```

Add comment

```
\chapter{An Introduction to Lua\TeX}
```

```
\section{What is it--and what makes it so different?}
```

Lua\TeX{} is a `\textit{toolkit}`--it contains sophisticated software tools and components with which you can construct (typeset) a wide range of documents. The sub-title of this article also poses two questions about Lua\TeX: What is it--and what makes it so different? The answer to “What is it?” may seem obvious: “It’s a `\TeX{}` typesetting engine!” Indeed it is, but a broader view, and one to which this author subscribes, is that Lua\TeX{} is an extremely versatile `\TeX`-based document construction and engineering system.

```
\subsection{Explaining Lua\TeX: Where to start?}
```

The goal of this first article on Lua\TeX{} is to offer a context for understanding what this TeX engine provides and why/how its design enables users to build/design/create a wide range of solutions to complex typesetting and design problems--perhaps also offering some degree of “future proofing”

```
\chapter{Lua\TeX: Background and history}
```

```
\section{Introduction}
```

Lua\TeX{} is, in `\TeX{}` terms, “the new kid on the block” despite having been in active development for over 10 years.

```
\subsection{Lua\TeX: Opening up \TeX's “black box”}
```

Knuth’s original `\TeX{}` program is the common ancestor of all modern `\TeX{}` engines in use today and Lua\TeX{} is, in effect, the latest evolutionary step: derived from the `pdf\TeX{}` program but with the addition of some powerful software components which bring a great deal of extra functionality.

```
\end{document}
```


L^AT_EX

1. How many built-in sectioning levels does LaTeX support?
2. Can you name them in order from top to bottom?



Resources worth checking out!

- [Harvard – LaTeX Resources on Overleaf](#)
 - [Overleaf – Bibliography Management in LaTeX](#)
 - [BibTeX.eu – natbib Reference Manual](#)
 - If you are able to review only one resource, this would be the most valuable
-
- This makes a great background listen—better than music, maybe.
 - <https://www.youtube.com/watch?v=emmBqI046Qg>
 - <https://www.youtube.com/watch?v=2BdBfsXbST8>
 - This episode of the [Lex Fridman](#) podcast features **Donald Knuth**, the creator of TeX and a pioneer in computer science.

Referencing: why and how?

- Don't just copy & paste
- Use your own voice
- Indicate that this is someone else's idea

